



Illuminating
ENGINEERING SOCIETY

APPROVED METHOD:
PHOTOMETRIC TESTING OF
SEARCHLIGHTS USING
INCANDESCENT OR HID SOURCES
AN AMERICAN NATIONAL STANDARD



ANSI/IES LM-11-20(R2023)

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AN AMERICAN NATIONAL STANDARD**

Publication of this Committee Report
has been approved by the IES.
Suggestions for revisions should
be directed to the IES

**Prepared by
The IES Testing Procedures committee**



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Approved by the IES Standards Committee, December 19, 2022, as a Transaction of the Illuminating Engineering Society.

Approved April 28, 2023, as an American National Standard.

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Published by the Illuminating Engineering Society, 120 Wall Street, New York, New York 10005.

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Printed in the United States of America.

ISBN# 978-0-87995-012-5

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Foreword

The original guide for the “Photometric Testing of Searchlights” was started in 1951 and published in March of 1958. The Searchlight Committee and the Committee on Testing Procedures for Illumination Characteristics, along with T. H. Projector of the National Bureau of Standards, were instrumental in developing the final draft of that first report.

In general, this guide follows the same format but replaces the 1958 version and represents a revision to the 1997 guide. Main changes include: 1) an update of the references, including the addition of IES LM-75 for detailed information on goniophotometer types and photometric coordinates; 2) an update of the examples of typical searchlight systems; and 3) the introduction of the International Commission on Illumination (CIE) value f_1 as a measure of the acceptability of the spectral response of a photodetector used in measurement.

1.0 Objective and Scope

This Approved Method applies to lighting equipment having a total field angle of less than ten degrees (refer to **Normative Reference 2.1**). (**Section 6.6** provides an example for obtaining test data on field angle.) This includes equipment in which light is controlled by reflectors, lenses, or their combinations. The document applies to projectors with a variety of light sources, including tungsten filament, tungsten-halogen, and high-intensity discharge lamps. It does not apply to projector systems composed of arrays of light emitting diodes. For luminaires where the total field spread is equal to or exceeds ten degrees, the reader is referred to ANSI/IES LM-35-20.¹ Photometric testing of reflector-type lamps is described in ANSI/IES LM-20-20.²

2.0 Normative References

2.1 ANSI/IES LS-1-20

Illuminating Engineering Society. Nomenclature and Definitions for Illuminating Engineering. New York: IES; 2020. Available online: www.ies.org/standards/definitions/. (Accessed 2019 Jun 28).

2.2 ANSI/IES LM-45-20

Illuminating Engineering Society. IES Approved Method for the Electrical and Photometric Measurement of General Service Incandescent Filament Lamps. New York: IES; 2020.

2.3 ANSI/IES LM-51-20

Illuminating Engineering Society. Approved Method: Electrical and Photometric Measurements of High Intensity Discharge Lamps. New York: IES; 2020.

2.4 ANSI/IES LM-54-20

Illuminating Engineering Society. IES Guide to Lamp Seasoning. New York: IES; 2020.

3.0 Nomenclature and Definitions

The terms used in this document follow the definitions given in **Normative Reference 2.1**. Additional terms are defined below.

3.1 auxiliary equipment

The ballast and the igniter needed for the proper operation of discharge lamps.

3.2 units

The units of electrical measurement used in this Approved Method are the volt, the ampere, and the watt. The units of photometric measurement are the lumen and the candela. Color is specified in terms of the CIE recommended systems.

4.0 Physical and Environmental Test Conditions

4.1 General

Changes in ambient temperature or air movement have little influence on the electrical and photometric characteristics of incandescent filament and HID lamps. In the case of HID lamps, the discharge tube is encapsulated inside an outer bulb, which provides a level of thermal and airflow isolation. Thus, the luminous

flux remains almost constant over a wide range of ambient temperatures. It is good laboratory practice for the storage and testing of lamps to be undertaken in a relatively clean environment. For lamps that operate at a high bulb temperature, contaminants can be etched into glass bulb surfaces during operation. Therefore, lamps should be cleaned before measurement.

4.2 Temperature

The ambient temperature of the photometric laboratory shall be maintained at $25\text{ }^{\circ}\text{C} \pm 5\text{ }^{\circ}\text{C}$ ($77\text{ }^{\circ}\text{F} \pm 9\text{ }^{\circ}\text{F}$). The ambient temperature shall be measured at the same height and not more than 1.0 meter (3.3 feet) from the luminaire. Temperature sensors shall be shielded from direct radiation and convection currents produced by the luminaire.

4.3 Airflow

Airflow is necessary to avoid thermal stratification and to control the ambient temperature. Good thermal control requires a high number of air exchanges per hour in the test area, but should be accomplished with low airflow rates.

4.4 Stray Light

Stray light should be suppressed in the test environment through the adequate use of low-reflectance surface finishes, shielding, and baffling. The walls, ceiling, and floor of the photometric test room should be painted flat black or covered with a low reflectance matte material such as black felt, black velvet or black carpeting.

Opaque low reflectance baffles with limiting apertures interposed between the photodetector position and the luminaire position that shield the photodetector, except in the direction of the test source, should aid in suppressing the stray light reaching the photodetector.

Stray light may be measured by running a complete or partial test with all direct light from the lamp or luminaire shielded from the photodetector.

4.5 Vibration

Lamps should not be subjected to excessive vibration or shock during measurement.

5.0 Electrical Test Conditions

5.1 Power Supply Requirements

5.1.1 Primary Source Voltage. In the case of HID lamps, ballasts should be operated at their nominal rated primary voltage and frequency. If the voltage rating is a range, the ballast should be operated at the intended application voltage, and the test voltage shall be recorded in the test report.

5.1.2 Power Supply Wave Shape. The AC power supply, during the operation of the luminaire, shall have a sinusoidal voltage wave shape such that the value of the root mean square (RMS) summation of the harmonic components shall not exceed 3% of the fundamental. It is important to exercise caution when using constant voltage transformer regulators. This type of system can produce harmonic distortion in excess of 3% when lightly loaded.

5.1.3 Power Supply Regulation. During testing, lamp wattage should be regulated to stabilize light output. Lamp wattage control is achieved via power supply regulation. The method used depends on the type of ballast and lamp used for testing. For incandescent lamps, lamp wattage is controlled via current regulation. For HID lamps running on electronic ballasts, the ballast electronics act as a wattage regulator, and only the supply voltage is regulated during testing. For HID lamps running on magnetic ballasts, wattage is regulated via supply voltage regulation. In all cases, to ensure stable luminous output from the lamp, the regulated parameter, either current or voltage, should be tightly regulated. (See **Section 6.6.3** for regulation details for each lamp type.)

5.2 Reference Circuit Requirements

There are no reference circuit requirements.

5.3 Measurement Instrument Requirements

5.3.1 Electrical Instruments. Electrical parameters (e.g., voltage, amperage, wattage) shall be measured with instruments having National Metrological Institute (NMI) traceable calibration. The instrument measurement tolerance of AC voltage and current shall be less than or

equal to 0.2%. The instrument measurement tolerance of AC power shall be less than or equal to 0.5%. The instrument measurement tolerance of DC voltage and current shall be less than or equal to 0.1%.

Instrument measurement ranges should be selected so that the mid-to-maximum portion of the range is used for any specific measurement. Since gaseous discharge lamps may have seriously distorted wave shapes, AC instruments (for voltage, amperage, and wattage) shall measure true RMS values. Instruments with scales calibrated in RMS values but with deflections or readings based upon average or peak values shall not be used. If analog instrumentation is used, large-scale deflection for the intended measurement range should be used.

The electrical instruments should be chosen according to the guidelines and precautions presented in *ANSI/IES LM-28-20, Approved Method: Guide for the Selection, Care and Use of Electrical Instruments in the Photometric Laboratory*.³ It is important to note that some HID electronic ballasts provide a high frequency lamp current. Care should be taken to ensure that possible conducted or radiated electromagnetic interference (EMI) from this type of ballast does not cause reading errors. High frequency wire runs should be as short as possible and, if necessary, shielded from EMI. It may be necessary to use precautions in order to prevent EMI from reaching voltage monitoring equipment or the photodetector readout device through the power circuits.

6.0 Testing Procedure Requirements

6.1 General Procedure

In general, testing of photometric searchlight equipment should consist of^{4,5}:

- Determination of the requirements for, setup, and use of proper range facilities
- Selection of testing apparatus
- Selection and preparation of the equipment to be tested
- Procedure for taking the measurements
- Computation of test results
- Reporting of data

6.1.1 Goniophotometers. The usual method for measuring angles in the luminous intensity distribution measurements is to mount the test equipment on a goniophotometer, keeping the photometric equipment fixed.⁶ Basically, goniophotometers provide a means for mounting the equipment, for rotating it around the two axes (horizontal and vertical), and for measuring the angles of rotation. (See **Figure 6-1** and **Figure 6-2**.)

When the searchlight is of unusual size and weight it may be necessary to utilize its own mounting and goniometric facilities for the photometric work or to hold the searchlight fixed and traverse the beam by moving the detector.

6.1.1.1 Type A Goniophotometer. The polar axis of angular measurements will be vertical (see **Figure 6-1**) when the “horizontal” rotation causes the vertical axis of the projector to be tilted. For this condition, the revolving table is within the frame that provides the projector mounting and the lateral rotation of the projector about its vertical axis.

6.1.1.2 Type B Goniophotometer. Searchlights may also be tested in such a manner that the polar axis of angular measurements is horizontal (see **Figure 6-2**) when the yoke of the goniophotometer is rotated horizontally about a fixed vertical axis. For this condition, the revolving table is outside of the yoke frame within which the projector is mounted.

6.1.1.3 Coordinate Systems. The grid lines shown in **Figure 6-1** and **Figure 6-2** represent spatial coordinate systems (photometric webs) around the luminaire, which is at the center of the sphere. ANSI/IES LM-75-19 provides detailed information on coordinate systems.⁶ Airport rotating beacons, which are basically searchlights, are measured with Type A coordinates, as these coordinates are used to define the location of aircraft relative to airport marker lights (and therefore in the beam pattern). Type A goniophotometry (as illustrated in **Figure 6-1**) is the preferred method for searchlight measurements, to ensure consistency between applications.

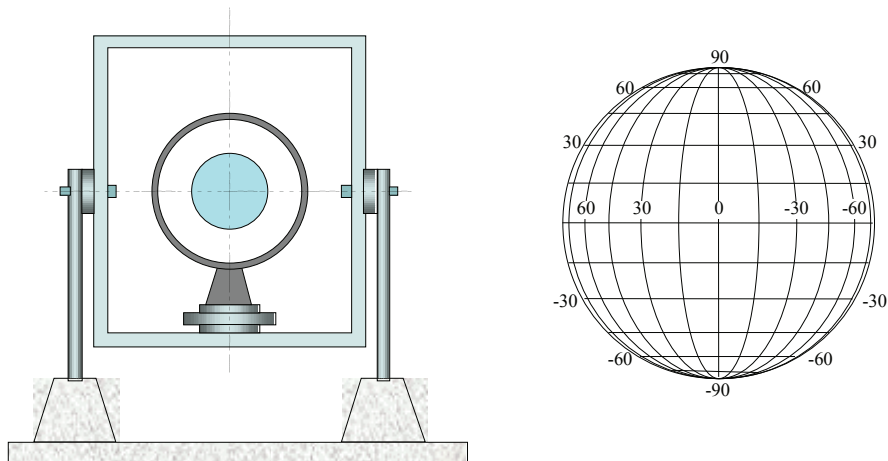


Figure 6-1. Type A goniophotometer, in which the polar axis is vertical. The photometric web, which represents the loci traced by the photodetector as the goniophotometer axes are rotated, is generated by step increments about the horizontal axis followed by a full rotation of the vertical axis. The full range of the vertical axis is -90 to +90 degrees, while the range of the horizontal axis is -180 to +180 degrees. (© Illuminating Engineering Society)

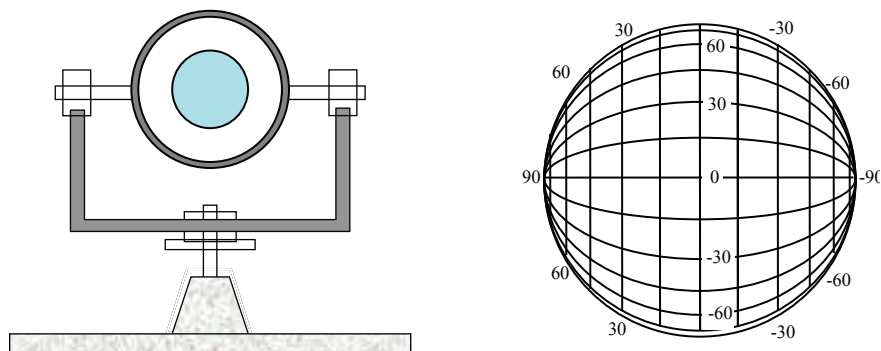


Figure 6-2. Type B goniophotometer, in which the polar axis is horizontal. The photometric web, which represents the loci traced by the photodetector as the goniophotometer axes are rotated, is generated by a step increment of the vertical axis followed by a full rotation of the horizontal axis. The full range of the horizontal angles is -90 to +90 degrees, while the range of the vertical angles is -180 to +180 degrees. (© Illuminating Engineering Society)

6.1.1.4 Photometers for Searchlight Measurements. The equipment required for photometric measurements of searchlights is basically the same as that required for other types of photometric measurements.^{7,8} Variations in ambient temperature on outdoor ranges, however, may adversely affect the performance of the photometer, and thus increase measurement uncertainty. The use of insulated and thermostatically controlled housings is recommended for areas prone to large temperature fluctuations.

The photometer head, or photodetector, of the goniophotometer should have relative spectral responsivity matched to the $V(\lambda)$ function. The f_1' value of the spectral responsivity shall be less than 3%. For absolute measurements on spectrally structured sources (e.g., RGB or monochromatic), a systematic spectral correction factor should be considered. To determine such a correction factor, the spectral responsivity of the photometer head and the spectral distribution of the source to be measured should be known. It is

important to note that there are photometer heads readily available that have an f_1' value of spectral responsivity less than 1.5%.

Photocurrent in silicon photodetectors is typically linear over a dynamic range of approximately 10^6 . The absolute range for any given device is a function of the active area of the device and the electrical properties, such as shunt resistance, of the photodiode. The range of linear output for the device is typically defined as the range over which the photocurrent is linear within 1% and is significantly below the “saturation current” for the device. Care should be taken to ensure that the device is used within its linear range for the expected range of luminous intensity for the photometric measurements.

The calibration records for the photometric system should include the spectral response values of the detector from 380 to 780 nm, in increments no greater than 5 nm. The calibration documentation should also include the f_1' value.

6.1.1.5 Angular Positioning. Since searchlights may have a total field angle of less than one degree and may furthermore be of unusual weight, the mechanical requirements on the goniophotometer are severe. The system should have suitable rigidity, freedom from backlash, and low uncertainty in angular measurements. The system should support angular settings of 0.1° or less, with an uncertainty of $\pm 2\%$ or better.

6.1.1.6 Repeatability of Light Readings. The goniophotometer should have good repeatability in light readings. There are many factors that contribute to uncertainty in light readings, and the goniophotometer should be evaluated to determine whether the system’s measurement uncertainty is consistent with the measurement needs of the lamp or luminaire under test. Some factors influencing the system uncertainty are:

- Positioning repeatability
- Accuracy of gain ratios in amplifier and scaling circuits
- Positioning accuracy
- Quantization error in A/D conversion
- Linearity in the A/D conversion

- Spectral responsivity of the photodetector vs. the source spectral properties

The uncertainty factors for any given goniophotometer depend on the design and construction of the instrument and the relative contributions of the factors appropriate for that system.

Historically, verification of the photometer involved using a luminous intensity standard lamp with a calibration traceable to SI units through a National Metrology Institute (NMI). With this method, it is desirable to verify the response of the photometer at illuminance levels comparable (in magnitude) to those levels that will be measured. Sector discs and/or filters may be used to permit verification of the photometer response at selected levels. The use of filters requires care to avoid errors arising from spectral selectivity.

When a reference projector is used to measure the atmospheric transmittance of the range, it may be desirable to use the same projector to verify the response of the photometer. This requires an absolute calibration of the reference projector, instead of the relative readings that are suitable for calculation of the atmospheric transmittance. Once the reference projector is calibrated, a single reading of the luminous intensity provides a correction factor for the photometric range as a whole (including the photometer response and the atmospheric transmittance over the length of the range). When the difference between the luminous intensity of the reference projector and the test projector is two orders of magnitude or greater, it is desirable to use a sector disc or some other means to equalize the illuminance levels incident on the photodetector produced by the two projectors.

A preferred method for verifying the response of the photometer involves using calibrated photometers rather than standard lamps.⁹ Adopting the detector-based method eliminates the need to maintain calibrated sources. Secondary calibrated photometers may be generated by placing their photodetectors on an optical bench on the same illuminated plane as a calibrated photodetector. This method is claimed to have considerably less uncertainty than the method using standard lamps. Calibration of the photodetector

shall be valid up to and above the highest value of illuminance that will be encountered during measurement of the searchlight beam.¹⁰ In many cases, this will include levels of illuminance of up to 100,000 lux (9,300 footcandles).

6.2 Range Requirements

For the purposes of determining the requirements for the proper range facility, it is assumed that the projector apparatus uses a parabolic mirror or Fresnel lens to collimate the light emitted from a source located at the focus of the optical system. Typical searchlights use a “shallow” parabolic mirror (wherein the radius of the aperture is less than twice the focal length of the mirror) with an incandescent lamp aligned perpendicular to the projector axis, or a “deep-dish” parabolic mirror (where the radius of the aperture is more than twice the focal length of the mirror) with either a xenon short-arc lamp, or compact tungsten-halogen lamp aligned with the projector axis. The use of high-intensity discharge lamps, especially compact metal halide lamps, is still limited. Projector systems using Fresnel lenses, such as the assembled lenses used in many lighthouses, are less common, but may be treated in the same fashion as projector systems using parabolic mirrors.

6.2.1 Range Length. If measured at distances greater than a certain minimum, the intensity of searchlights, beacons, and other highly collimating luminaires can be calculated by using the inverse-square law. The derivation of the minimum distance at which the inverse-square law holds is extensively discussed in Walsh.⁴ For a projector apparatus using a parabolic mirror, this minimum distance is a function of the focal length of the reflector (f), the radius of the projector aperture (R), and the radius of the smallest element of the light source, whether an arc stream or filament (r). This minimum distance point is called the beam crossover point and is the point where the optic is viewed as completely flashed. Only at distances greater than the beam crossover point does the inverse-square law apply. The minimum inverse-square distance for on-axis measurements, measured from the center of the projector aperture, may be calculated by using **Equation 6-1**, the general formula (see **Figure 6-3a**).

$$L_0 = 0.001 \left(\frac{Rd}{r} \right), \quad (6-1)$$

where:

L_0 = the distance to the beam crossover point, measured from the aperture plane of the optic (in meters)

R = radius of the projector aperture (in millimeters)

d = distance from the light source to the aperture limit (in millimeters)

r = radius of smallest element of light source, e.g., the arc stream width of an arc source or one coil of a multi-coil filament lamp (in millimeters)

Note: In Walsh,⁴ the crossover distance (L_0) is calculated from the base of the mirror, while in this document it is calculated from the aperture plane of the mirror. Thus, in Walsh the calculated value for L_0 is greater, by a factor of $R^2/4f$, than the calculation of L_0 shown herein.

In almost all cases, the source may be approximated as a sphere located at the focus of the projector optical system (see **Figure 6-3b**), and the formula in **Equation 6-2** may be used.

$$L_0 = 0.001 \left[\frac{R}{r} \left(f + \frac{R^2}{4f} \right) \right] \quad (6-2)$$

Equation 6-2 is valid for projector apparatus using either a shallow or deep-dish parabolic mirror.

For a projector apparatus with a shallow parabolic mirror and an incandescent lamp with a C-13 or similar “flat” filament arrangement, the filament may be treated as approximating a disc located at the focus of the projector optical system (see **Figure 6-3c**). The effective radius of the disc source is the radius of the projected area of the disc as viewed from the extreme edge of the optical system. The formula in **Equation 6-3** may be used.

$$L_0 = 0.001 \left[\frac{R}{r} \times \frac{\left(f + \frac{R^2}{4f} \right)^2}{\left(f - \frac{R^2}{4f} \right)} \right] \quad (6-3)$$

Equation 3 is not valid for systems where the radius of the projector aperture is equal to or greater than twice the focal length of the optical system. At this point, the projected area of the disc source becomes zero, and the minimum inverse-square distance goes to infinity.

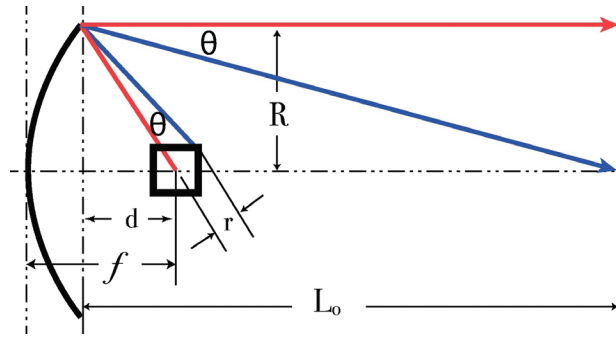


Figure 6-3a

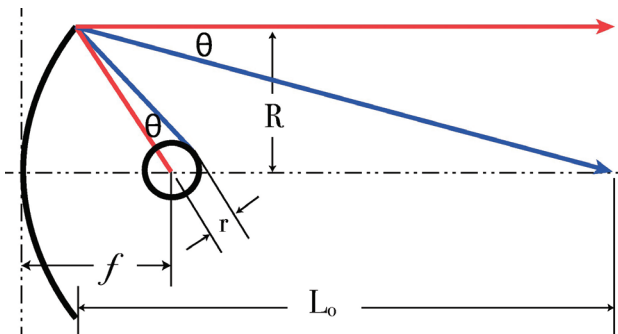


Figure 6-3b

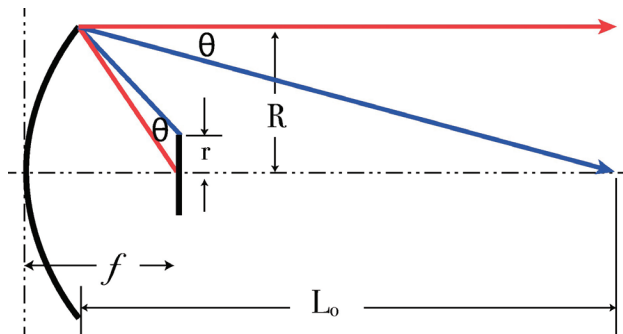


Figure 6-3c

Figure 6-3. Beyond a certain minimum distance, L_0 , the intensity of searchlights can be calculated using the inverse-square law. Calculation of this minimum distance is dependent upon variables such as the radius of the smallest element of the light source (arc width or filament), as shown in (a), and whether this light source can be treated as approximately spherical (b) or disc-shaped (c). (© Illuminating Engineering Society)

Equations 6-1 through **6-3** assume ideal light sources and optical components, and measurement of the luminous intensity on the axis of the projector apparatus. Therefore, the results of any calculations using these equations should be considered approximate. In addition, it is typically necessary to measure the distribution of the luminous intensity across the projected beam.

The distance at which the inverse-square law may be presumed to be valid for off-axis measurements is longer than for on-axis measurements. There is no convenient general solution to determining this distance exactly, but it may require an analysis of the beam structure for each particular case. Alternatively, a practical solution is to check the inverse-square law at two distances to see whether it holds reasonably well.

Methods for using shorter ranges and *zero-length photometry* have been devised; however, the full-length range is preferable for producing the least

uncertainty in measurement. These methods and a more complete discussion of minimum inverse-square distance calculations are available.^{4,5,11}

6.2.2 Alternative Short Test Distance Method. A method has been developed for testing searchlight distribution at short test distances. This method is designated the *Zero-Length Searchlight Photometry System*.^{11,12} The Warfare Vision Branch, Fort Belvoir, Virginia, developed a goniophotometer based on zero-length photometry. It used an $f/4$ 100-cm (40-inch) diameter parabolic collimator to accept and focus the searchlight beam to a sampling aperture. Comparing the measurements of the zero-length photometry system to results obtained at longer test distances indicated a variance of about 3%. A similar system, with an $f/10$ 91-cm (36-inch) diameter parabolic collimator, illustrated in **Figure 6-4**, was established at the U.S. Coast Guard Research and Development Center, Groton, Connecticut. In the U.S. Coast Guard system, the mirror

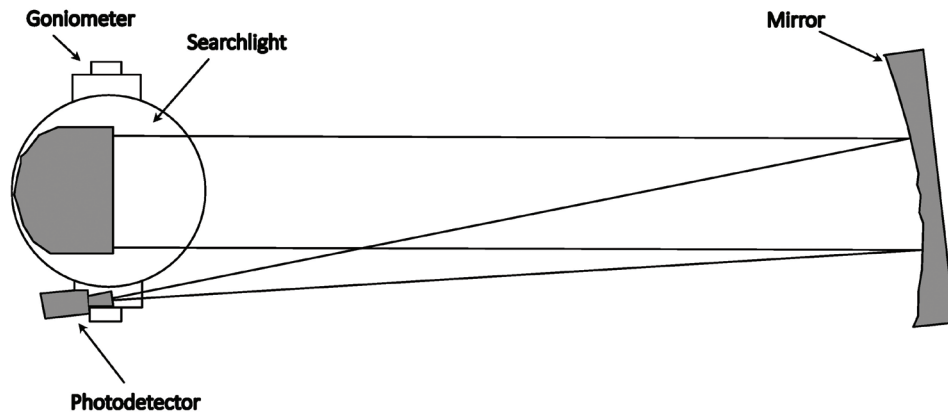


Figure 6-4. Measurement of a searchlight beam pattern is possible at short distances using a Zero-Length Searchlight Photometry System, such as one installed at the U.S. Coast Guard R&D Center. The photodetector is placed at the focus of the $f/10$ 91-cm (36-inch) parabolic mirror, which focuses collimated rays in the beam onto the photodetector aperture. (© Illuminating Engineering Society)

is sited at the end of the range so as to place the aperture of the photodetector, mounted on the frame of the goniophotometer, at the focal point.

6.3 Range Location

In general, indoor ranges are preferable but are frequently impractical because of the lengths required. For photometry on the relatively shorter indoor ranges, proper photometric procedures are essential in order to minimize uncertainty.⁴ Outdoor ranges suitable for daytime measurements require much more attention to methods of reducing stray light and minimizing the uncertainty of results due to atmospheric conditions. Range sites should be selected where the terrain is flat, and the range should be as high off the ground as practicable. Ranges should not be located where dust or moisture is prevalent. Stray light should be minimized by a suitable system of diaphragms. Any remaining stray light should be measured and subtracted from all readings.

6.4 Correction for Atmospheric Transmission

The absorption of light by moisture, smoke, and dust particles, even in an apparently clear atmosphere, may introduce considerable errors in measurements.¹³ It is therefore desirable to measure the atmospheric transmittance before and after the test has been made. A standard reference projector is frequently employed, and is calibrated either by repeated observations in the clearest weather (when the atmospheric transmittance

may be accurately estimated or independently measured), or by laboratory methods (see **Annex B**).

6.5 Selection and Preparation of the Equipment for Test

6.5.1 Selection. The searchlight being tested should represent a sample of the manufacturer's regular product. It should be checked to ensure that it is in good optical, mechanical, and electrical condition.

6.5.2 Mounting. The optical head of the searchlight should be mounted on the goniophotometer with the intersection of its axes of rotation close to the intersection of the axes of rotation of the goniophotometer. In the event that the projector does not have rotation gear, it should be mounted so that the center of the projector aperture is close to the intersection of the axes of rotation of the goniophotometer.

6.5.3 Focusing. Projectors should be focused to give the narrowest and most sharply defined beam pattern at approximately the same distance as that at which the test is to be made. Methods for focusing projectors for both a finite distance and for infinity are provided in **Annex C**.

6.5.4 Cleaning. Unless the effects of dirt accumulation are to be tested, all components of the optical system should be cleaned before testing.

6.6 Test Lamps

6.6.1 General. Lamps with a stable light output are needed for photometric test purposes and should remain so during a test or series of tests when constant line voltage is supplied. Test lamps should be selected for close conformance to the lamp manufacturer's design dimensions and construction.

6.6.2 Lamp Seasoning. Test lamps shall be seasoned such that their characteristics remain constant during conduction of the test. (See **Normative Reference 2.4**, *ANSI/IES-LM-54-20, Approved Method: IES Guide to Lamp Seasoning* for detailed information.)

High intensity discharge (HID) lamps require special precautions when operating orientation is changed. Long stabilization periods (up to several days) may be necessary to re-stabilize the lamps in the new orientation. Even designated "universal operating" HID lamps require prolonged stabilization in their actual operating position before being used as test lamps. In practice, separate lamps should be maintained for orientation (base up, base down, or horizontal) and should be operated only in their assigned orientation. Lamp seasoning should also be done in the same orientation as the intended orientation during testing.

6.6.3 Test Lamp Operation.

6.6.3.1 General. During photometric testing, stable lamp performance is of primary importance. The lamp is considered stabilized when the monitoring of light output over a period of 30 minutes produces fluctuations no greater than $\pm 1.0\%$ between any three consecutive readings taken approximately 15 minutes apart. Care shall be taken to ensure that successive light readings are not increasing or decreasing monotonically. A false sense of stabilization can occur if the sampling period is short compared to the normal lamp stabilization time.

6.6.3.2 Incandescent Filament Lamps. A stabilization time of 10 minutes is usually sufficient for incandescent and tungsten-halogen lamps. Tungsten-halogen lamps shall be operated at a constant value of current near their nominal voltage. Other incandescent filament lamps should be operated at a constant current near

90% of rated input lamp voltage (at approximately 75% of nominal luminous flux this practice extends the life of test lamps). Current for incandescent lamps shall be regulated within $\pm 0.1\%$ of nominal current (DC operation may be necessary to achieve this current regulation requirement). Operation at other voltages shall be specifically noted in the test report.

For more information on electrical measurements the reader is referred to **Normative Reference 2.2**, *ANSI/IES-LM-45-20/R23, Approved Method: Electrical and Photometric Measurement of General Service Incandescent Filament Lamps*.

6.6.3.3 HID and Short-Arc Lamps. A stabilization time may range from one to three hours for short-arc, mercury, high pressure sodium, and metal halide lamps. HID and short-arc lamps operating on magnetic ballasts shall be operated within $\pm 2.0\%$ of the rated power of the lamp and regulated to within $\pm 0.5\%$. Lamps operating on electronic ballasts shall have the ballast input voltage set to within $\pm 0.5\%$ of the unit's rated primary voltage and regulated to within $\pm 0.2\%$. If the rated primary voltage is a range, the test voltage shall be recorded in the test report. Metal halide lamps should not be moved after turn-off until the salts within the arc tube have cooled and solidified. For most applications, 30 minutes of cooling is sufficient.

For more information on electrical measurements the reader is referred to **Normative Reference 2.3**, *ANSI/IES-LM-51-20, Approved Method: Electrical and Photometric Measurements of High Intensity Discharge Lamps*.

6.7 Test Data

In general, testing of searchlights may involve:

- Horizontal and vertical traces of luminous intensity
- An isocandela diagram of the beam profile
- Luminous flux of the projected beam
- Efficiency of the light source

6.7.1 Horizontal and Vertical Traces of Luminous Intensity. The beam pattern is scanned about the horizontal and vertical planes that intersect along the optical axis of the beam pattern (see **Figure 6-5**). The scans should encompass a range that is at least twice the nominal field angle.

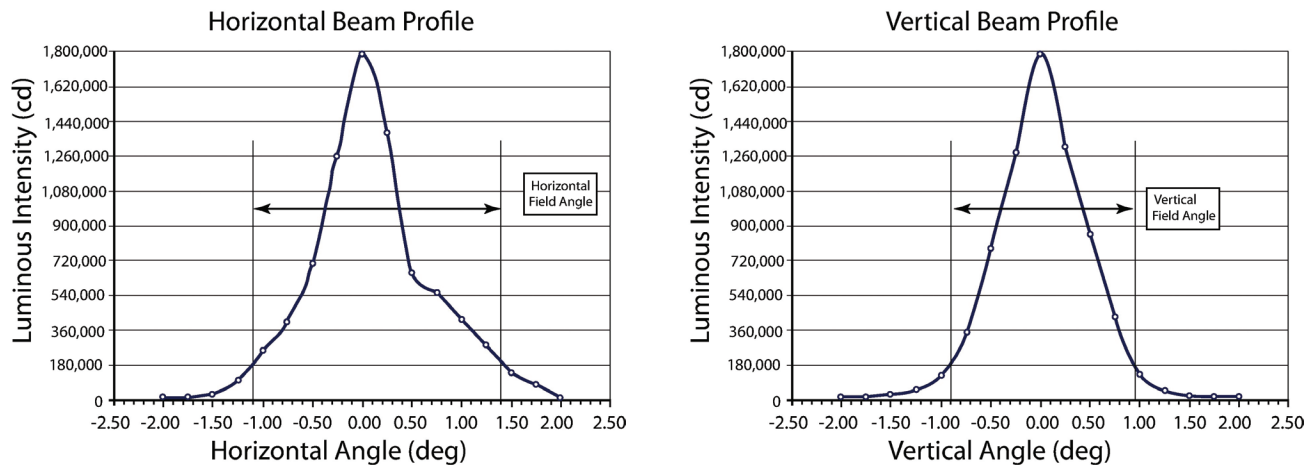


Figure 6-5. These luminous intensity distributions for a searchlight under test reveal the horizontal and vertical beam spread (divergence). (© Illuminating Engineering Society)

6.7.2 Iso-candela Diagram of the Beam Profile. A sufficient number of readings are taken to adequately determine the beam profile throughout the range of desired horizontal and vertical angles (see **Figure 6-6**). The horizontal and vertical extents should encompass an area that defines the beam throughout the field angle.

6.7.3 Luminous Flux of the Projected Beam. The luminous intensity values for each set of vertical and horizontal angles are used to calculate the luminous flux for the corresponding angular zones. The angular steps should be uniformly spaced, and should be sufficiently small such that a minimum of 100 readings are made within the limits of the beam, defined as the points at which the luminous intensity falls to 10% of the highest value within the beam (the solid angle described by the horizontal and vertical field angles). The luminous intensity values at each corner of a “cell” are averaged to produce the mean luminous intensity for the cell, which is used in the calculation. A cell in which the luminous intensity of any corner point equals or exceeds 10% of the highest value within the beam should be included in the calculation of the luminous flux of the projected beam. (See **Figure 6-6** and **Figure 6-7**). The searchlight data in **Figure 6-7** were obtained by taking uniformly spaced measurements to explore the entire intensity distribution at suitable angular steps. The calculated values of luminous intensity are converted into the luminous flux for the corresponding angular zones. It should be noted that horizontal and vertical “lumen

totals” are tallied to the right of and along the bottom of the table. The flux computed for the portion of the distribution lying outside the field may then be added to that lying within the field to obtain the total flux. (For this example, a step size of 0.25 degrees, both vertical and horizontal, was selected for ease of presentation. A smaller step size should be used for actual calculation, to ensure a minimum of 100 individual measurements within the defined limits of the beam.)

The total lumen output of a projector beam may be measured directly by projecting the beam into a large integrating device that has been adapted for such measurements.^{3,14}

6.7.4 Efficiency of the Light Source. To fully analyze the performance of a searchlight system and calculate changes in the emitted beam (both peak intensity and field angles) due to changes in the light source, it may be necessary to determine the distribution of the source’s luminance as it is projected into the collecting angle of the searchlight optical system.

6.8 Computation of Results

6.8.1 Computation. The luminous intensity of each angular position within the searchlight beam is computed from the measurement of the illuminance on the photometer detector by means of the following formula:

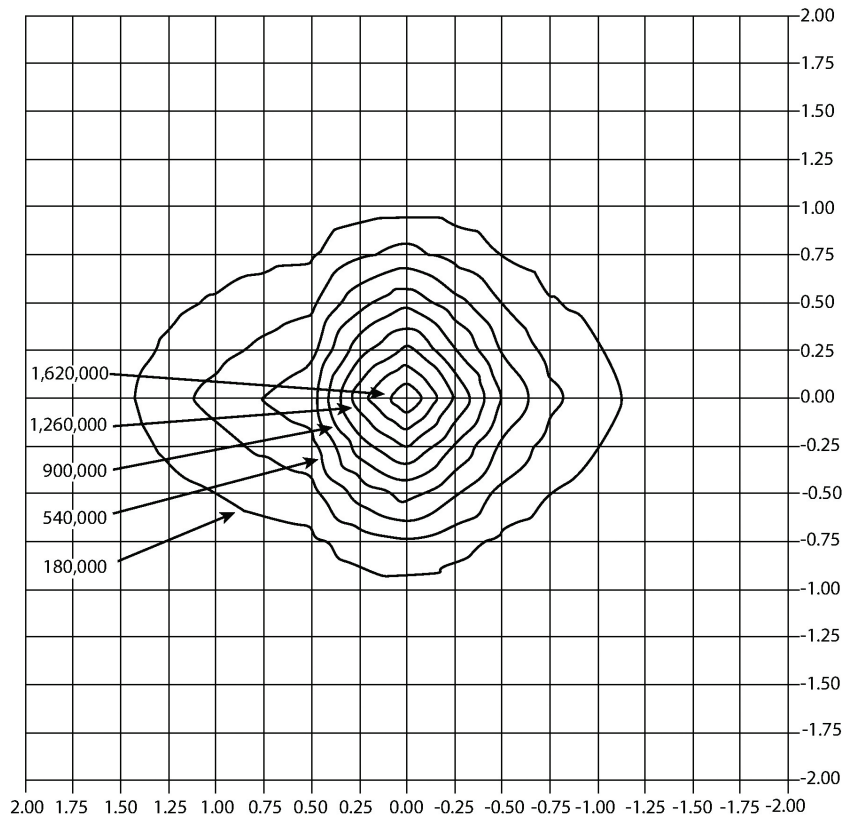


Figure 6-6. An iso-candela diagram is expressed as continuous lines connecting points of equal luminous intensity within the beam pattern. A sufficient number of readings are taken to adequately determine the beam profile through the range of horizontal and vertical angles. (© Illuminating Engineering Society)

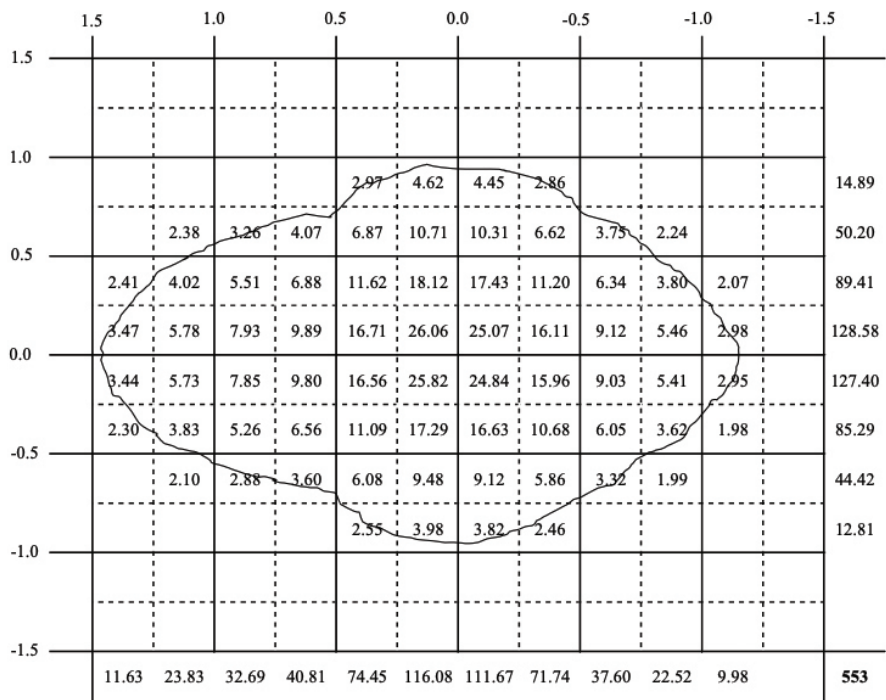


Figure 6-7. An example of test data for a searchlight. (© Illuminating Engineering Society)

$$I = \frac{(E_1 - E_2) \times D^2}{T^D}, \quad (6-4)$$

where:

- I = the luminous intensity of the searchlight
- E_1 = total illuminance measured by the detector
- E_2 = portion of the total illuminance due to stray light
- D = the test distance
- T = atmospheric transmittance per unit distance (see **Annex B** and **Reference 12**)

The term E_2 in **Equation 6-4** is for stray light that reaches the detector by means other than directly from the projector under test. *Stray light control* includes preventing light that is emitted by the projector, and reflects from walls, floors, and other surfaces, from reaching the photodetector. *Ambient light control* includes eliminating or reducing the amount of light from sources other than the projector under test. The impact of ambient light may be determined by removing power to the projector and recording the output of the photodetector. The impact of both elements may be determined by taking measurements with the projector on, but with the direct light path occluded. Normal practice is to block the direct light by means of a curtain or other barrier and then take a reading. Some laboratories make a complete scan for stray and ambient light and then program the computer to subtract an appropriate amount from each recorded value of the measurement scan.

If the light source is operated below its rated lumen output, the scaling of the intensity shall be corrected accordingly.

6.8.2 Process. The process of developing useful results may involve the following steps:

1. Plotting horizontal and vertical candela distributions, and from these, if required, deriving iso-candela diagrams
2. Computing the luminous flux in each individual measurement cell (see **Section 6.7**) using the proper zonal solid angular area in steradians (refer to ANSI/IES LS-4-20, *Lighting Science: Measurement of Light—The Science of Photometry*¹⁵);

Note: In general, care should be exercised when using zonal areas, such that the angular arguments conform to the type of goniophotometer employed (see **Section 6.1.1** and **Reference 6** in particular). However, in the case of searchlights as defined in this guide (i.e., projectors with field angles [divergences] of less than 10 degrees), the errors in beam flux calculations that will occur if the wrong coordinate system is used are small (less than 1%¹⁶). This error is avoidable. Converting the angles to the correct set of axes should be completed before lumen calculations are made.

3. Summing the lumens of each individual measurement cell, using the procedures outlined in **Section 6.7**.
4. Computing the beam efficiency of the searchlight by dividing beam lumens by total lumen output of the test lamp.

7.0 Reporting of Data

Figures 6-6 and **6-7** are examples of results contained in typical test reports. The test report should also include the following information, where applicable:

- Projector description, including:
 - Manufacturer's name
 - Type and catalog number
 - Dimensions
 - Size and shape of the optical components
 - Dimensions and optical characteristics of the cover glass
 - Description of any auxiliary optics (if used)
 - Other essential information, including the method of selecting the sample
- A description of the light source, including:
 - Lamp manufacturer and manufacturer's designation
 - Lamp type (e.g., tungsten-halogen, xenon short-arc, metal-halide)
 - Rating (watts, volts, or amperes)

- Rated luminous flux
- Luminance or luminance distribution
- Bulb type
- Base type
- Normal burning position
- Light center length (LCL)
- Filament construction or arc tube size
- Actual light source dimensions
- Light center position during test
- Maximum luminous intensity of the beam (in candelas)
- Field angle, in degrees, in both the vertical and horizontal planes
- Beam angle (full-width at half maximum), in degrees, in both the vertical and horizontal planes
- Iso-candela diagram or candela distribution curves
- Lumen distribution in the beam
- Total luminous flux within the beam—beam lumens
- The beam efficiency of the searchlight system
- The distance at which measurements were made—test distance
- Date(s) of the test(s)
- Notes of any modifications in the preceding suggested method

Annex A — Inverse-Squared Distance Calculations

Sample calculations of minimum inverse-square distances are shown in the following two examples.

Example 1:

It is proposed to determine the minimum range length for photometric measurements on a xenon short-arc searchlight with a plasma ball approximately 2 mm in diameter, and a reflector with an aperture diameter of

355 mm (14 in.) and a focal length of 60 mm (2.4 in). The plasma ball of the xenon short-arc may be considered a spherical source located at the focus of the mirror. Using **Equation 6-2** in **Section 6.2.1** (spherical source) yields:

$$L_0 = \left[\frac{177.5}{1} \times \left(60 + \frac{177.5^2}{4 \times 60} \right) \right] / 1,000 \cong [177.5 (60 + 131)] / 1,000 \cong 34 \text{ meters}$$

This calculation is based on ideal conditions, and for on-axis measurement. In order to accommodate small imperfections in the construction of the optics and filament, it will be desirable to select a longer test distance. In this case, a test distance of 50 meters may be appropriate.

Example 2:

A searchlight with a BTR 1,000-watt tungsten-halogen lamp, which has a filament that is approximately 12.5 mm (0.5 in) square in size, has a reflector with an aperture diameter of 362 mm (14.25 in.) and a focal length of 102 mm (4 in.). The filament may be approximated as a disc with a radius of 6.35 mm (0.25 in.). Using **Equation 6-3** in **Section 6.2.1** (disc source) yields:

$$L_0 = \left[\frac{181}{6.25} \times \frac{\left(102 + \frac{181^2}{4 \times 102} \right)^2}{\left(102 - \frac{181^2}{4 \times 102} \right)} \right] / 1,000 \cong 44.3 \text{ meters}$$

This calculation is based on ideal conditions, and for on-axis measurement. In order to accommodate small imperfections in the construction of the optics and filament, it will be desirable to select a longer test distance. In this case, a test distance of 75 meters may be appropriate.

Annex B — Atmospheric Transmittance Determination

B.1 General

The atmosphere absorbs and scatters light, sometimes to an extent that can significantly reduce the illuminance on a detector over the typical distances of a searchlight range. It is therefore necessary to measure the transmittance of the range and make

appropriate corrections in the photometric readings. Atmospheric transmittance conditions can vary; thus, periodic checks of range transmittance should be made and test readings corrected accordingly.

B.2 Methods

Three basic methods of determining atmospheric transmittance have been satisfactorily used and are recommended. They are: 1) reference projector method; 2) two-station method; and 3) target luminance method.

B.2.1 Reference Projector Method. Essentially, this method uses a projector whose minimum inverse-square distance is short enough to permit measurement in an indoor range and yet has sufficient luminous intensity to permit a satisfactory reading on the photometer at the end of the test range.

The reference projector is mounted in place of, or in close proximity to, the test unit. The beam is then projected over the entire length of the range as used for the test unit. It is, therefore, possible to use this method to establish a correction factor for the detector and range. If atmospheric correction alone is required, only relative readings (illuminance at the time of the test compared to illuminance at 100 percent transmittance) are required.

The reference projector should employ a stable light source and be accurately aligned with the detector. Low voltage, high wattage incandescent filament lamps can be used and operated below their rated voltage to ensure stability. To ensure consistent results, the projector should have a lightly etched cover glass to smooth out the filament images and give a uniform luminous intensity over a relatively wide angle from the beam center.^{17,18}

B.2.2 Two-Station Method. This method consists of measuring the illuminance and computing the apparent luminous intensity of a reference projector at two different distances along the range to determine the loss in intensity, which represents the loss due to atmospheric transmissivity over the range. Under this method, calibration is not necessary for either the reference projector or the photometer to determine the transmissivity over the length of the range.¹⁹ It

is necessary that both measuring points be located beyond the minimum inverse-square distance for the reference projector. It is important to note that either the reference projector or the photodetector may be moved to create the two measurement distances. The response of the photometer shall be linear across the different levels of illuminance that will be measured across the two stations. If different photodetectors are used in this method, their response curves shall be verified relative to each other.

The reference searchlight shall have a stable light source and be accurately aligned with the detector(s). Low voltage, high wattage incandescent filament lamps can be used and operated below their rated voltage to ensure stability. To ensure consistent results, the projector should have a lightly etched cover glass to smooth out the filament images and give a uniform distribution over a relatively wide angle from the beam center.

The computation of the transmittance of the range with the two-station method is based on Allard's law:

$$E = \frac{I \times T^D}{D^2}, \quad (\text{B-1})$$

where E is the illuminance at a distance D from a source of intensity I when T is the transmissivity or transmittance per unit distance.

The formula to be used in computation of transmissivity is:

$$\bar{T}_{D_1} = (E_1 D_1^2 / E_2 D_2^2)^{[D_1 / (D_1 - D_2)]} \quad (\text{B-2})$$

where:

T_{D_1} = the transmittance of the range ($= T^{D_1}$);

D_1 = the length of the range

D_2 = the distance of the second station

E_1 = the measured illuminance at D1

E_2 = the measured illuminance at D2

A sample computation of range transmittance can now be made. A test range 408 meters (1,340 feet) long is assumed, with a second station at 45 meters (150 feet), ascertained to be beyond the minimum inverse-square

distance. It is also assumed that at a given time, E_2 is 95 times as great as E_1 . Applying the formula for range transmittance:

$$\overline{T}_{D_1} = (1 \times 408^2 / 95 \times 45^2)^{[408/408-45]} = 0.85 \quad (\text{B-3})$$

The transmittance over the range is 85 percent. It is convenient to prepare a table or a graph showing the relation between the ratio $E_1:E_2$ and the range transmittance. It is important to note that the two-station method assumes that the atmospheric transmissivity is uniform over the length of the range. This is not necessarily true; therefore, care should be used when employing this method.

B.2.3 Target Luminance Method. Another method of measuring transmission is based on the use of a telephotometer to measure the luminance of a target surface over the length of the range.^{20,21} The target shall be of sufficient size to fill the aperture of the telephotometer when viewed from the far end of the range and shall have a uniform luminance. The luminance of the target is first measured at a relatively close range and then again from the end of the range. Use of a telephotometer with a variable aperture will allow for measurement of the same area of the target, thereby reducing the uncertainty due to non-uniformity of the luminance. The ratio of the apparent luminance as measured on the range to the luminance as measured at a close distance is taken to be the transmittance over the range. For daytime use, a self-luminous target (emitting surface) shall be used. Measurement of the target luminance, with the target source turned-off, will allow for subtraction of the ambient luminance from the luminance measured when the target is energized, yielding the values of target luminance used to calculate the transmittance over the range. The target luminance should be at least two orders of magnitude greater than the luminance of the target surface under ambient lighting in order to minimize the uncertainty of the calculated transmittance.

Annex C — Focusing Methods

C.1 General

The manufacturer's instructions for aligning and focusing the searchlight should be followed. Typically, these instructions will result in a system that is focused at infinity. In the event that no instructions are available, or when the searchlight needs to be focused at a shorter distance (i.e., a test distance that is not appreciably greater than the minimum inverse-square distance), then the methods listed herein may be used. If the searchlight is focused for a specified test distance, then it should be refocused to infinity after photometric measurements are completed.

Section C.2 describes methods of aligning the light source along the optical axis of the mirror. **Sections C.3** through **C.6** provide methods of focusing the searchlight at a defined test distance, while **Sections C.7** and **C.8** provide methods of focusing the searchlight for infinity.

C.2 Visual Alignment

Prior to focusing, it is necessary to check or adjust the alignment of the light source on the optical axis of the searchlight. For a searchlight with a shallow parabolic mirror and a filament lamp, one method for determining the axis of the searchlight, and for locating the light source upon it, is to stretch a pair of strings across the aperture of the reflector so that their intersection lies at the center of the aperture. Looking through the lamp bulb toward the reflector, there will appear, in order, the lamp filament, the crossed strings, an image of the crossed strings, and an image of the filament. (The lamp may be turned on at low voltage to facilitate observation.) If the center of the filament, the intersection of the strings, and their images all lie in the same straight line, then the lamp is on the optical axis. This does not ensure that the lamp filament is at the focal point of the mirror. Any departure of the geometric axis of the aperture from the optical axis of the reflector will result in a proportional displacement of the images of the strings from the strings. This method cannot typically be performed with a deep-dish parabolic mirror, as the source is generally aligned along the optical axis.

A second method of visual alignment may be conducted by projecting four parallel laser beams onto the searchlight mirror. The beams may be generated by individual lasers, or through the use of beam splitters and reflectors. The laser beams are typically arranged so that the pattern formed by the beams is uniformly distributed about the optical axis of the mirror. While that arrangement is not absolutely necessary, it may be helpful in ensuring the alignment of the beams relative to each other. The beams should converge at the focal point of the mirror. Thus, this visual alignment method will also provide for focusing the searchlight for infinity. Personnel using this method should follow proper safety procedures while working with lasers.

C.3 Visual Adjustment

In this method, the beam is projected onto a target normal to the beam axis and a distance from the projector that is approximately equal to the test distance. The target should be large enough to include the major portion of the beam, and the projector is focused until the narrowest and most sharply defined beam pattern is obtained. This method is satisfactory for indoor ranges and for relatively short outdoor ranges when the focusing can be done at night.

C.4 Adjustment for Optic Flash

If a uniform source is placed at the focal point of a perfect optic, and the optic is observed from a point on axis at or beyond the minimum inverse-square distance, the aperture of the projector will appear as a uniformly luminous disk. The optic is then said to be “fully flashed.” It is, therefore, possible to determine the correct focus of the projector for a given distance by adjusting the position of the source along the optical axis until the aperture appears to be emitting with a uniform luminance. The average luminance of the aperture should also be the maximum value obtained at the prescribed distance. It is necessary that the optic be observed from a point along the beam axis and at the distance at which measurements are to be made.

In practice, optics are imperfect and light sources are more or less non-uniform, so that it is often difficult to obtain uniform flashing of the optic aperture. Furthermore, some obstruction is caused by placement of the lamp and lamp holder within the projector system. If the lamp

is small relative to the optic, any displacement of the lamp from the focal point will result in incomplete or non-uniform flashing. In addition, the minimum inverse-square distance may be quite large, resulting in the need to view the optic through a telescope. In practice, it may be more desirable to use this method only as a check for both alignment and focus.

C.5 Formation of Image of Distant Target at Focal Point

If a small card is held at the focal point of the projector optic, an image of a distant illuminated target may be seen upon it; the image will be small and distorted, but if the central portion of the image is as sharply focused as possible, the projector is in focus. Because of the small size of the image, it is desirable to use a fairly large target of high contrast with the background.

C.6 Trial Luminous Intensity Distributions

Another method of obtaining exact focus and optimum results is to take a series of short luminous intensity distributions through the approximate axis of the beam, each distribution being taken after a small change in the focal adjustment. The best focus may be determined as that giving the distribution with the maximum luminous intensity and minimum beam spread. This focusing method is recommended in all cases where it is practical. With multi-coil filament lamps some caution is needed in applying this method since the luminous intensity distribution may contain several more or less well-defined peaks from filament images.

C.7 Perforated Disc and Screen Focusing

Essentially, this method takes a number of small portions of the beam at predetermined locations and adjusts the focusing mechanism until these miniature beams are parallel to each other and to the optic axis. Any number of small beam portions may be taken, but in practice, the number needed should be relatively few.

An opaque shield of a size to cover the entire beam is mounted at the opening of the projector in such a manner that it is adjustable with respect to the projector. Five holes—approximately 1.5 millimeters (approx. 1/16 inch) in diameter—are drilled in this shield, spaced in such a manner that one will be on the optic axis and the other four located 90 degrees apart on a radius

that is equal to two-thirds the radius of the reflector. A moveable screen should be mounted at a distance approximately 5 to 20 times the aperture diameter. On this screen are located five points that correspond exactly to the five holes in the opaque shield. The center hole is located on the optical axis, and the face of the screen is perpendicular to this axis when the projector is leveled so that this axis is horizontal.

The light source will produce five sharp images on the screen. The center hole will produce two images if the optic is a reflector: one by direct light and one by reflected light. The image produced by direct light will be larger and not as well defined as the image produced by reflected light. The focusing mechanism is adjusted until the two images from the center hole in the shield are superimposed upon each other and all images fall upon their respective points on the screen. The degree of coincidence will depend upon the accuracy of the reflector.

If an auxiliary reflector is used, a secondary set of images will be produced. The main adjustment should be completed as just described, with the auxiliary reflector removed from the searchlight. The auxiliary reflector is then installed and adjusted by alignment of the secondary images in the same manner as the primary adjustment was made, without disturbing the primary adjustments. With the auxiliary reflector in place, the images formed by the center hole will be occluded. The adjustment should be made so as to build up dark areas (e.g., areas between filament coils of the primary images) with the secondary images.

C.8 Projected Shadow Method

This method²² is useful when the light source is small or when a small light source may be substituted conveniently for the searchlight's source. It is based on the principle that the shadow of an obstruction placed in a perfectly collimated beam corresponds exactly to the cross-section of the obstruction. Since a defining condition of the best focus of the light source in a searchlight is that the projected beam is most nearly collimated, a comparison of the projected shadow of an obstruction in the beam with the obstruction itself may be used to make the adjustment for best focus. Any of a variety of obstruction masks may be used, as determined by convenience and the optics in the searchlight.

One convenient form for the mask is a hollow square cut from a metal sheet. The diagonal of the outer edge is slightly shorter than the diameter of the optic. The linear dimensions of the inner edge are about two-thirds to three-quarters of the dimensions of the outer edge. An exact outline of the edges of the mask is traced on a screen, which should be placed at a distance from the aperture plane of the optic that is twice the aperture diameter. The mask is placed close to the aperture plane. The planes of the mask and of the screen are made parallel with the aperture plane of the optic. If the searchlight's light source is not small enough to cast a fairly well defined shadow of the mask on the screen, it should be replaced by a smaller source, such as an ophthalmic lamp. The light source is then moved in and out along the optic axis until the position is found at which the edges of the projected shadow most nearly lie on the traced outline, thus establishing the position of best focus.

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Order# ANSI/IES LM-11-20(R2023)

ISBN# 978-0-87995-012-5

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